

ACTIVITY 5

LEVELS:

Grade 12, University Preparation

COURSES:

Canada and World Issues
HEV.01, HEV.02, HE1.01, HE1.03,
HE2.05

TASK:

Through a simulation exercise,
Students work in small groups
to learn about resource
sustainability.

EXPECTATIONS:

The student will:

- Analyse the impact of resource consumption patterns and evaluate the effects on people and environments at the local, national, and global level.
- Explain how people perceive resources and sustainable development differently at different times and in different places.
- Evaluate some of the ways of promoting sustainable consumption patterns and assess their effectiveness in selected places and regions of the world.

TIMEFRAME:

30 Minutes

ACHIEVEMENT CATEGORIES:

Knowledge / Understanding
Thinking / Inquiry
Communication
Application / Making Connections

MATERIALS:

Bag of Bean Seeds; Paper and pencils for keeping score; Worksheets 7 to 10
Suggested Homework 2; flipchart paper and markers

VOCABULARY:

Renewable resource
Environmental carrying Capacity
Per capita
Sustainability



Sharing the Wealth

PROCEDURE:

1. Divide the group into communities of four.
1. Hand out one set of guidelines to each group (See Worksheet 7).
2. Ask each group to come up with a name for their community.
3. Place 16 beans in a communal pile for each community. Instead of beans you could use beads or buttons.
4. Explain the rules of the game to your students:
 - The beans represent a valuable renewable resource. The resource is replenished after each round of play.
 - Each community member is given a role card that will tell them how many beans they take from the resource pile each round (See Worksheet 8).
 - Each community member must take at least one bean in each round to survive.
 - One person in each group must record the number of pieces taken by each community member in each round (See Worksheet 9).
5. After each round, count how many beans each community has remaining in the pile, and add an equivalent number of beans to the pile.
6. Play three or four rounds, pausing after each round to find out how many community members did not survive.
7. Play one final round and then record how many of your community members survived. Collect the beans and replace them in the bag.
8. Return to your communities of four to play an extension of the game you just finished.
9. Place 16 beans in a communal pile for each community.
10. Explain the rules of the game:
 - You are an indigenous community of Nicaragua. You depend on the land in your community to grow food. The jelly beans represent the produce from growing food. Each year, you find that your garden is producing less. The soil is dry and erosion is taking place.
 - Similar to the original game, the resource is replenished after each round of play, except that this time, only half as many beans are replenished as what remains.
 - Each community member is given a role card that will tell you how many beans they take from the resource pile each round (See Worksheet 10).
 - Each community member must take at least one bean in each round to survive.
11. One person in each group must record the number of pieces taken by each community member in each round.
12. After each round, count how many beans each community has remaining in the pile, and add half that number of candies to the pile.
13. Play two or three rounds, pausing after each round to find out how many community members did not survive.
14. Play one final round, then have community members share what happened in their communities:
 - Compare the results to the previous round. Use the tabulation sheet in Activity Sheet 9 to compare your findings.
 - Use the Discussion questions on the next page as a guide for your discussion.

DISCUSSION COMMUNICATION

- Which communities are confident they will have enough candy for everyone as long as the pile is renewed? How did these communities arrive at that point? What strategies were used?
- Was there a leader in these communities? If so, why did the community listen to that person?

DISCUSSION ON SHARING THE WEALTH

KNOWLEDGE / UNDERSTANDING

1. In which communities did everyone survive? Compare the survival rate between the two games.
2. Could these communities have reached “sustainability” without communication?
3. Which communities had the most beans in the resource pile at the end of the game? Compare the quantity remaining between the two games.
4. Compare per capita ownership around the room. Out of the whole room, who had amassed the most beans? Compare the per capita ownership between the two games. How did he/she accomplish this?
5. Did this keep others from surviving?

APPLICATION / MAKING CONNECTIONS

6. Where do we see this kind of greed in the real world?

THINKING / INQUIRY

Start a discussion of the following:

7. What information is necessary to know how to manage a resource sustainably? (e.g. community size, resource renewal rate, environmental carrying capacity, etc.)
8. What is needed to put information into practice (e.g. leadership, communication, trust, legislation, understanding of consequences, examples of failure, etc.)?
9. As a member of a Third World community, what kind of actions can you take to improve your situation (e.g. form a community group to help manage resources sustainably, practice conservation methods and environmentally sound farming practices, etc.).

APPLICATION / MAKING CONNECTIONS

10. What other resources can you think of that could be threatened by unsustainable use?

⁹ McKeown, Rosalyn. Ed. 2002. Adapted from “Drain or Sustain?” in Education for Sustainable Development. <http://www.esdtoolkit.org>

HOMEWORK SUGGESTION (See Suggested Homework 2)

Select a resource from the list below:

Clean Water



Fertile Soil



Clean Air



1. How is this resource used?
2. What activities threaten its sustainability?
3. What measures can be taken to preserve this resource?